



How to Support Rapid Prototyping of Virtual Content for Automotive AR Head-Up Displays?

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Abstract. Emerging AR applications for vehicles, and especially for AR HUDs, are current topics for modern assistance systems to support drivers in their driving tasks. Prototyping of AR experiences is presented in current research in immersive authoring tools to overcome complexities and entry barriers for designers and psychologists. However, these are not applicable to navigation displays for AR HUD. This paper presents a qualitative study with concept developers in their domain to determine the characteristics of AR applications that the HUD should possess. The different dimensions of the problems and properties of the required information are presented. The different facets of associated issues and information requirements are also discussed. These facets and its corresponding information are categorized in different levels of complexity of the navigation context, which are not typically part of the development of standard AR applications. Technological entry barriers introduce additional complexity levels for the concept developers. More specifically, the representation of dynamic virtual objects on displays in dynamic environments complicates the development as there is no suitable evaluation platform meeting the requirements of the concept developers.

Keywords: Augmented Reality · AR HUD · Development process · HMI prototyping

1 Introduction

Augmented Reality (AR) refers to virtual information used to supplement the user's view of the real world [2]. In the automotive domain, AR is used to present assisting virtual information within the driver's field of view, such as navigational information, street signs and speed limits. The aim is to reduce the driver's mental workload associated with the need to combine information from multiple sources in the vehicle (e.g. speed indicator) and on the road. Abdi and Meddeb demonstrated that reducing mental workload could lead to less driving errors and therefore in less traffic accidents [1]. Bauerfeind et al. and Sandbrink showed that due to the ever-growing number of vehicles and the resulting traffic load, using

AR as assisting technology in the vehicle will become increasingly important [3, 16]. Therefore, novel prototyping methods and tools are required in order to create supportive and compelling AR visualizations for use in vehicles.

1.1 Development of AR Applications

AR applications exist in different forms:

- Handheld AR: A device such as a tablet PC or smartphone is held by at least one hand. A live camera video is displayed on the device’s screen and virtual content is merged into the video feed.
- Head-worn AR: Users wear see-through devices (e.g. a Microsoft HoloLens) on their head. They see the real-world environment combined with spatially registered 3D content that is superimposed on the environment within their line of sight.
- Projection-based AR: The position of user’s head is tracked. A static or movable projector projects additional content on the surrounding environment, resulting in an AR view [11].

In this work, we refer to the examples above as *standard AR applications*. They apply different kinds of displays (monitor, projector) to present spatially registered 3D content. Users see the content and can interact with it, in which case the content reacts to the users’ commands in real-time (Fig. 1).

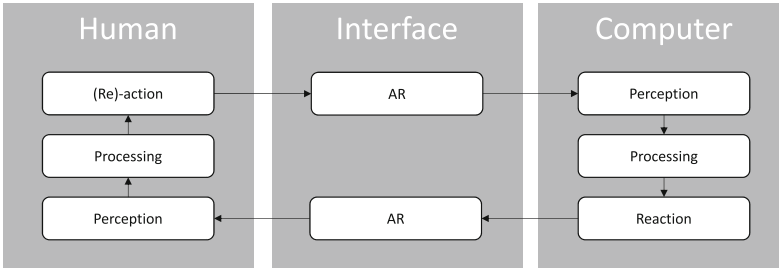


Fig. 1. Augmented Reality serves as the interface between human and computer.

The development steps for standard AR applications are known and already supported by numerous frameworks. Software developers, UX designers, and application specialists usually work in teams. The planned target platform defines the necessary SDKs for the development. 3D objects and their behaviors are created and imported. One important step is the spatial mapping of the 3D data to a world coordinate system. This process may need spatial markers, model-based or feature-based tracking approaches and specific mapping procedures. Specialized AR-relevant workflows and authoring tools such as Unity

MARS¹ can be used if the physical space of the users is unknown to the developers. The application will then be deployed to the target platform and gradually tested, refined and published upon completion.

The development of assisting AR applications in the automotive domain can require the consultation from psychologists and human factors experts in addition to the standard developer team. Since the use of AR in the car has the potential to distract the driver, the cognitive perception of the displayed content can be investigated and explored in a joint effort by designers and psychologists. Such procedure could lead to significantly more development iterations compared to standard AR applications. In addition, some technical requirements of SDKs created for release on certain AR platforms may not be as available and applicable in the vehicle. In standard AR applications it is usually sufficient to use in-built sensors to recognize simple surfaces, building or room structures to map augmented content. In the vehicle, the mapping process can be a lot more complex: AR content must be adapted not only to the user's position and dynamics but also to vehicle dynamics and dynamic changes in the environment.

It is not unusual that new AR concepts are developed while the target vehicle platform is not yet available, making the implementation of such AR concepts extremely challenging. To our knowledge, no public SDKs exist to facilitate the AR software development of in-vehicle Head-Up-Displays (HUDs). Therefore, any development is done on prototype setups or in simulated virtual vehicles. This can lead to additional iterations in the product development cycle, introducing undesirable delays and conflicts between hardware architecture requirements and user experience.

1.2 Prototyping and Evaluation of AR HUD Concepts

Crucial AR HUD development phases must be iterated several times (see Sect. 1.1).

1. Ideation: Creative concept development by discussing and brainstorming ideas.
2. Prototyping: Idea and concept visualization.
3. Development: Software implementation including definition of behaviors, trigger events, markers etc. of the final experience.
4. Evaluation/Testing: Testing the implemented concept in a simulator and/or in the real driving experience.
5. Presentation: Final concept presentation depending on the project dependencies and environments.

According to Mueller and Thoring, the prototyping phase is characterized by a visualization of ideas prior to their implementation in a finished product [13]. Such tangible visualization allows assessment, testing and refinement at a low cost. Nebeling and Madier as well as Rudd et al. distinguish between low fidelity

¹ <https://unity.com/products/unity-mars>.

to high fidelity prototyping [14, 15]. Low fidelity prototyping is a simple and low cost method to quickly represent ideas. The skills related entry barrier is low allowing individuals from different professional backgrounds to participate in the prototyping process. However, low fidelity prototypes comprise limited functions and interaction techniques. A low fidelity technique like paper prototyping in AR lacks immersion, does not provide a spatial view or a 3D-prototype. High fidelity prototypes on the other hand can include the full range of functions and be fully interactive, which is associated with higher costs and a longer implementation time. As an AR-prototyping technique, high fidelity prototyping offers specific advantages [5, 7, 12]:

- It provides a high level of immersion allowing the user to experience the virtual content as seamlessly embedded into the real world.
- It ensures higher ecological validity, increasing the extent to which findings from test setups can be applied to real-world situations.

To achieve a high fidelity of AR prototypes for automotive AR HUDs, expert level skills in 3D modelling, behavior programming and rendering as well as knowledge of specialized software (e.g. Blender and Unity 3D) are required. Often, this part of development is completed by experts instead of AR concept creators.

1.3 Motivation and Goals

Concept developers of automotive AR HUD visualizations cannot rely solely on low- or medium-fidelity prototyping. They would greatly profit from being able to work with a high fidelity prototyping environment considering its higher ecological validity. Therefore, the following study was designed to gain a deeper understanding of the automotive AR HUD concept development process by focusing on the following research questions:

- Needs and requirements: Why do concept developers want a high fidelity prototyping environment? What needs should be met by a high fidelity prototyping environment?
- Pain points, hurdles and entry barriers: What keeps concept developers from working with a high fidelity prototyping environment? How can these challenges be overcome?

2 Related Work

In 2020, Freitas et al. reviewed literature dealing with tools and techniques for AR prototyping and ranked them according to the following categories [7]:

1. Resulting prototype fidelity: the level of similarity between the resulting prototype and the final product ranging from high to low. For example, the paper prototyping technique generates a low fidelity prototype whereas prototyping in Unity produces a high fidelity prototype.

2. Required skill level: skill level required to use the tool in accordance with the resulting prototype fidelity. For example, paper prototyping is ranked low in both categories, since sketching on paper in 2D requires does not require specialized expert skills, while prototyping in Unity is ranked highest in both categories.

Depending on the context and the desired results, developers may have to consider a trade-off between prototype fidelity, fidelity in AR and the necessary skill level when choosing an appropriate AR prototyping tool/technique.

At best, an AR prototyping tool or technique should allow a high prototype fidelity, be easy to use for individuals of variable skill levels while also producing a high fidelity in AR. Typically, this combination is rare and difficult to achieve, since producing high fidelity in AR is challenging without expert skills. Freitas et al. cite the following tools/techniques that fulfill the aforementioned criteria [7]. GestureWiz developed by Speicher and Nebeling is a tool for gesture-based interaction with virtual objects [17]. The tool was realized via the so-called Wizard of Oz approach, in which a person (the wizard) recognizes the user's gestures and mirrors them allowing users to realistically prototype gestures for interaction with virtual objects. With PintAR by Gasques et al. users can digitally sketch their ideas on handhelds in 2D, allowing an easy creation without expert skills [8]. Using Head-mounted displays (HMDs) users can view and interact with their prototypes in 3D with high levels of immersion. Using Holobuilder by Speicher et al. users can easily create AR-based instruction manuals for industrial applications [18]. According to Freitas et al., the Holobuilder tool is similar to PowerPoint [7].

In their user study, De Sá and Churchill implemented a mobile AR application as a high, low and mixed fidelity prototype [6]. Their results showed that the mixed fidelity approach was sufficient in this use case. Interestingly, users expected a higher fidelity of the high fidelity prototype than it could deliver, which could be a source of irritation and introduce additional challenges during the implementation process.

According to Freitas et al., PapAR by Lauber et al. is a technique that produces a low prototype fidelity and a low level of fidelity in AR [7, 10]. Since no expert skills are required, ideas can be easily expressed and visualized. With PapAR, users prototype on two layers separate layers. This approach is particularly suitable for AR, since one layer can represent the real world and the transparent layer can be used to add AR objects to the scene. While the PapAR technique retains the low skill level requirements, it increases the ecological validity of traditional paper prototyping. FrameBox and MirrorBox by Broy et al., increase the fidelity level by expanding paper prototyping with a third spatial dimension and don't require expert skills to use them [4]. Both Lauber et al. and Broy et al. have applied their tools and techniques to prototype infotainment systems in automotive HUDs [4, 10].

3 Use Case and Implementation

To find out how AR concept developers work, the study implemented a use case where participants developed an AR concept for an AR HUD. Therefore each participant was given a Lego City package containing a street layout, traffic lights, and street signs. With the help of two toy cars, the scenario was a traffic situation of a left turn with an oncoming car. This specific situation is complex since it required the participants to orchestrate two cars about traffic regulations, traffic lights, and street signs. Additionally, the participants were provided Play-Doh in different colors for prototyping 3D objects. It allows shaping the objects in any way. A laptop stand helped the study director observe the whole setup from the best angle. The setup (see Fig. 2) was neither validated nor used before. The study took a qualitative approach in which participants' rule

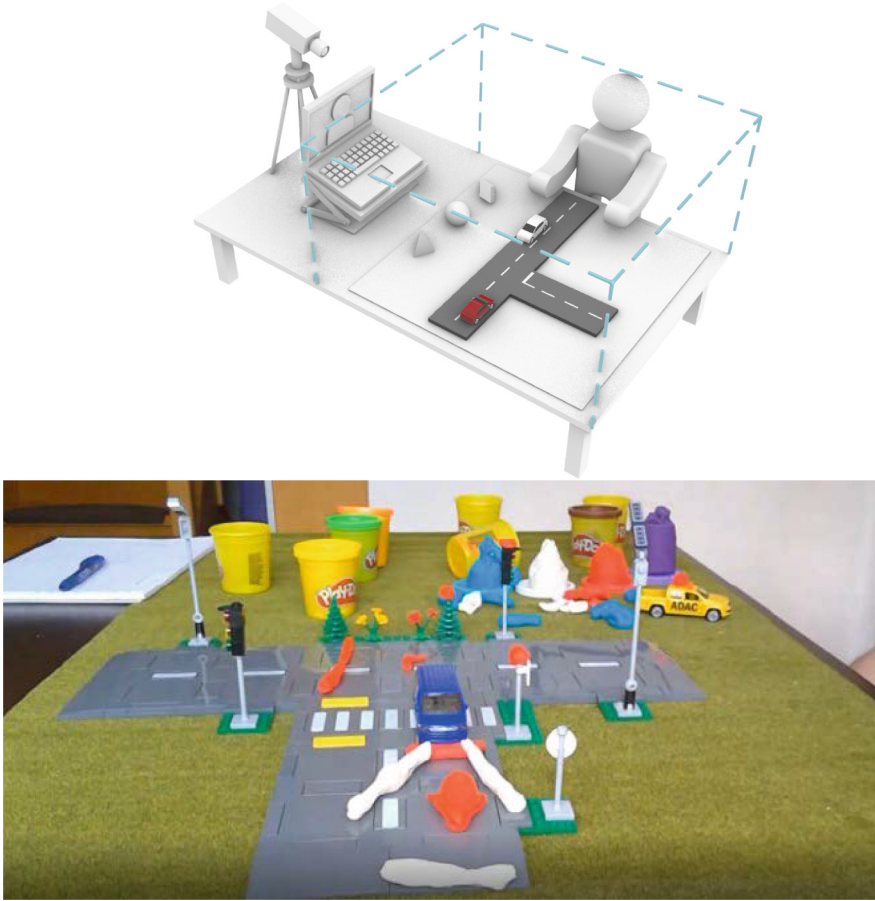


Fig. 2. Study setup - planned setup draft (top) actual setup from a participant (bottom)

sets for development and thought processes were evaluated using a systematic content analysis of video recordings and interviews.

With the help of the traffic scenario and Play-Doh, the concept developer could visualize his thoughts during the interview, supporting the process's creativity. The study aimed at observing the participants while completing a given task and gaining insights into their very own development processes concerning:

- What and where are pain points and hurdles for concept developers in their daily work?
- Which methodical and technical aid do concept developers wish for if they would get a HMI that would help them with their problems?

This refers to the guiding research questions presented in Sect. 1.3 and were the structuring part of the conducted interview.

4 Procedure and Evaluation

The user study took place in the first half of 2021. Due to the Covid-19 pandemic it had to be done remotely. $N = 8$ participants from automotive industry took part without compensation. All of the deliberately chosen participants were AR concept experts working with AR for a minimum of three years and daily. Here, we call them *concept developers*.

Three of them were psychologists, and five were either visual or interaction designers. Three participants held a Ph.D. degree, and the others had a master's degree. It is important to note here that these concept developers usually have technical knowledge, regardless of their specific education. The participants are experts in their field to create concepts, but they do not have all the necessary technical skills implement AR concepts in the vehicle themselves. In total, the study took around three to four hours per participant. Additionally, the participants received their working materials in home office.

With the help of the study director, who was connected remotely via video camera, they set up the traffic scenario with the two toy cars (see Fig. 2). Based on the setup, the study director explained the study in detail. The procedure depicted an exemplary development process divided into the phases described in Sect. 1.2.

The evaluation or testing phase was not part of the study because the participants were to be asked at what point in the development process they would conduct an assessment. Thus, the evaluation questions have moved on to the other phases. The development and presentation phases were classified as interview phases because the participants answered them hypothetically. Finally, ideation and prototyping were classified as working phases since they required the participants to work with the setup, for example, kneading 3D objects from play-doh. Here, the participants first had some time to fulfill the task and think aloud while the study director was observing. Afterward, the study director conducted the interview, depending on whether the questions were answered beforehand.

In addition to general questions, he asked contextually and phase-specific questions, such as questions regarding the experience and a theoretical HMI supporting the participants in the respective phase. Next, each phase is described in detail (Fig. 3):

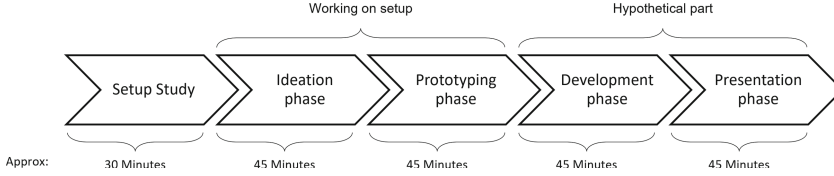


Fig. 3. Procedure of the study

1. **Ideation:** Participants had 15 min to develop possible traffic scenarios and how to support the driver in this situation in the given task. Afterwards, the interview was conducted. The questions mainly revolved around a theoretical HMI to support the participants during the ideation process.
2. **Prototyping:** The prototyping phase proceeded analogue to the previous phase. The participants used their time to specifically think about the implementation of the navigation concepts from the previous ideation phase. The questions mainly revolved around a theoretical HMI to support the participants during the prototyping process. Additionally, the interviewer asked questions regarding the implementation of virtual objects.
3. **Development:** The development phase was conducted as an interview only. The interviewer asked questions about the 3D modelling process, the implementation of virtual objects and behavior programming.
4. **Presentation/Transfer:** The presentation/transfer phase was conducted as an interview only. The interviewer mainly asked questions about the use case presentation. Lastly, participants were interviewed about other possible use cases, issues and hurdles experienced during the development of navigation concepts for AR HUDs.

The study was evaluated according to Kuckartz' structured content analysis [9]. The main thematic categories were derived from the research questions described in Sect. 1.3. The transcripts were coded using the thematic categories. Subsequently, all text lines with the same codes were compiled into a matrix. New subcategories were inductively formed based on the compiled matrix. In a next step, the resulting matrix was re-coded using the differentiated category system summarized in Table 1.

Table 1. Summary of the differentiated category system

User experience	AR authoring	Mobility functions
• Ease of use • Object modelling • Annotation notes	• 2D-Tools • Primitives • Tangible prototyping • Scenarios and components • Version management	• Timing and procedures • Information loss • Driver perspective • Spatial contextualization • Special features

5 Results

Every participant described a complete theoretical prototyping HMI for the end customer and could distinguish between basic terms and elements like virtual objects, objects related to the traffic scenario and behavior of the virtual and street elements. Table 2 shows a summarized overview of the main outcomes. In some study runs, it had become apparent that the ideation and prototyping phases could not be separated, as the participants had already began to integrate virtual content and prototype while still in the ideation phase. Therefore, it was essential for them to work with the virtual content at an early stage and specify their thoughts in the spatial context of the task.

Table 2. Overview of the main outcomes of the specific phases during the study

Working phase	Ideation phase	Prototyping phase	All phases
Completion	Naming of features	Casual and spatial description	Willingness to experience in real environment
Heat map		Driver-HUD-relation	
Mental model		Relation to daily work	
		Spatial reference	
		Description of behavior	

The evaluation of the working parts showed that almost every participant pointed out significant areas where information should be displayed. The participants named them maneuver or stopping points (shown in Fig. 4). In general, it is important to point out that every participant can be characterized as one of the following two types of concept developers: the ones who describe their concept in a more structure-controlled and those who express their idea in a more sequence-controlled way. This influenced their designed concepts and how they explained them. The difference became especially noticeable during the traffic events, like a traffic light that turns red. The structure-controlled developer recognized elements that could trigger an event and proactively described the event occurring. The sequence-controlled developer usually only paid attention to possible events during the maneuver or the route and inevitably brought the car

into an event. Thus, it happened four times that concept developers who took a sequence-controlled approach did not name the elements that could have been triggered in the scene. In the following, the two types shall be called structure and sequence developers. Table 3 presents each types properties.

Table 3. Structure and sequence developers

Developer	Property	Description
Structure	Description of user story	Possible events around the car
	Type of reaction to events	Proactive
	Note	Needed more complex information
Sequence	Description of user story	Linear on maneuver
	Type of reaction to events	Reactive
	Note	Were faster but ignored information

Figure 4 shows a compilation of the maneuver points of the scene, where AR content could be presented. In particular, it has been demonstrated here that the augmentation must take place far in front of the vehicle. The concept developer wants to control the timing of the display concerning vehicle position and speed. External triggers, such as traffic lights or crosswalks, are independent and can be displayed in regard to their importance. This figure is a compilation of the experts. Not all of them required the use of maneuver points. In comparison, others have explicitly marked some spatially in the scene.

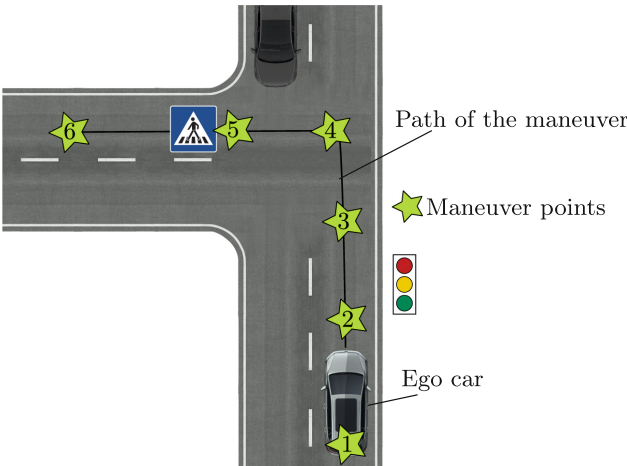


Fig. 4. Maneuver points (extracted from the videos of the study)

The ideation phase has shown that the concept developers, in particular, have indirect requirements for a development environment that do not necessarily have to be technical. One participant, in particular, emphasized that he needed the opportunity to collaborate with others and work on the concepts.

“[...] I need to collaborate with other persons, so multiple people need to see things together simultaneously. [...] whenever you work with AR and so on, especially the HUD, [...], even when you make studies, either you or the participant can see it. [...]”

The summarized results of the prototyping phase are shown in Table 4. They were assigned to the respective subcategories that emerged during the evaluation.

Table 4. Results of the prototyping phase

Subcategory	Summarized result
Description of all created virtual objects with Play-Doh	They have a spatial or causal relationship. Spatial is for registration of objects in the world coordinate system of reality, and causal is concerning road users, functional entities (such as traffic lights), or the driver himself The type of object generated can be described as single objects that change their shape concerning the emerging information. Another concept has been described as a permanent transformation of things throughout the maneuver so that recurring patterns can be generated
Relation between HUD and driver	The augmentation must always be one step ahead for the driver to react to the information in time The concept must also not overwhelm the driver. It must leave him a decision not to be forced to follow it
Comparison of the work in the study and daily work	Working with virtual dough would be very good. The spatial and tangible creation of virtual objects pleases almost everyone in the study setup 2D graphics are almost always created in daily work extended to three-dimensional space
Location of objects in scene	These depend on speed, distance to the driver, and the current amount of information in the display
Description of the behavior in general terms	The behavior was described based on emergent maneuver elements in the study setup They should be similar to the animation functionalities in Microsoft PowerPoint Common behaviors mentioned and used in the majority are changing the size of the virtual object, fading in and out based on distance, and pulsing

In the development phase, critical elements of development were asked. The summary shows aspects of the daily work of the concept developers. For example, Fig. 5 shows that half of the test persons are confident to prototype the concept created in the study, whereas 7 out of 8 would commission the development in their everyday work. This is due to time constraints, lack of competence in terms of knowledge about the car, or the ability to develop the concept with high visual quality for a presentation.

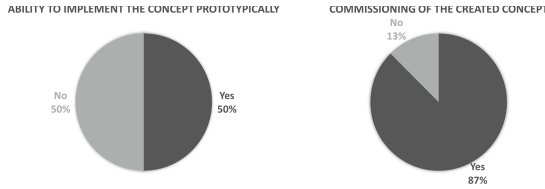


Fig. 5. Comparison of results in development

The effort was as high or very high by seven of the eight subjects (see Fig. 6). However, the evaluation was complex since they have a corresponding assignment in their daily work and can only estimate the actual effort. Whether prototypical or not, the development is a black box that the software developers implement.

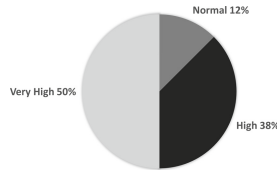


Fig. 6. Effort of programming

In the final phase of the study, only one participant communicated to show the customer's result in an AR HUD. The responses after the previous experience were characterized by alternative or no solutions. The following quote shows this: "[...] usually we can't present the product's final state. [...]". A large number was characterized by partial solutions, which can be attributed to two features of the presentation:

1. Only partial solutions are presented in partial functionalities based on triggers in simulated worlds and not under real circumstances.
2. The target group very much determines the presentation platform, and videos or presentations of moving images are shown, which, however, can represent the entire concept. The target audience includes potential customers, the general public, testers with specific characteristics or management.

In addition to the core results of the four phases, the participants were also asked about their willingness to evaluate the experience at this point, i.e., with the result generated in each stage. Table 2 shows partial statements of the respective phase.

Table 5. Willigness to evaluate

Phase	Quote
Ideation	“That avoids possible dead ends that you might only realize much later in the project if you skip this immersion phase in reality” “So it is necessary to test the whole thing at an early stage, so that you don’t realize after two years of development that the concept works with one person, but not with the other”
Prototyping	“[...] In a real situation it feels different than just in a micro world, like here now. I will understand something different than you. In my head something completely different happens than in yours and above all [...] also something completely different happens than in that of the future users”
Development	“Yep and on almost daily basis really. that’s critical at that point” “Only at the end, I wouldn’t want to see every little change I make in the real conditions but when I say ok I am done with this phase then I would want to experience it. [...] In the earlier phases it was more important to experience earlier because you were testing a concept, here it is at the end to confirm that everything works”

At the end of the study, the participants were asked to name the most significant problem from their point of view. The following summarized list shows the result:

- spatial understanding of the user are the biggest learning costs of AR
- to get into presentation mode
- to setup study settings
- mandatory reference to the environment combined with the limitation of the field of view
- create ideas in the ideation phase
- transition between the phases: media change or further tools needed
- getting good quality and stable data
 - occlusion is still a big problem
 - precise localization of the car and the environment are hard to process.

In summary, the statements on the last question are a good example of the different focal points and dimensions that concept developers in the automotive industry deal with as so called non-technical experts. In essence, as with the usual AR development process, the focus is on technical problems. They become apparent by naming technical difficulties in development and transferring results

between different media or platforms. Further on, procedural difficulties also become apparent, presented by a lack of cooperation in the development of virtual objects in the context of navigation for an AR HUD. At several points in the interviews, the desire for a collaboration environment was expressed.

6 Discussion

The results of the expert study essentially show how multidimensional the development of AR applications is in the automotive context. Half of all test persons had the confidence to prototype their personally created concept, which shows that the necessary technical basis exists to understand and name the general problems. In many places, however, clear so-called “black boxes” were called in which concept developers do not know what is happening or how the concepts are implemented. Therefore, in addition to strong design and programming skills, technical understanding of the vehicle and platform-specific development skills for the AR HUD are also needed.

This is particularly evident in the individual platform decisions made in day-to-day work. No holistic evaluation platform exists that meets the needs of concept developers. In most cases, partial solutions are created that are tested in simulation environments concerning experience or functionality. This clarifies that although all concept developers develop navigation concepts for AR HUDs, they have individual development processes about their decisions.

Indeed, there is a difference in whether a proof of concept is evaluated on slides, videos, simulated virtual worlds, or the actual target platform. This is shown by the statements in Table 5 regarding immersive evaluation. The study design allowed participants to manage their analyses and tasks spatially and tangibly. Even without an explicit question within the interview, almost all participants still positively mentioned it. Especially in the first two phases of the development process, the subjects’ spatial representation of all components and interrelationships helped.

As expected, the lo-fi prototyping environment used in the study helped the participants to understand the development process better. Even though the methodology is not a validated approach, it can be used as a first creative development building block in many places. No technical skills are necessary, and it is multi-user capable. This means that hardware setup can be used for simple virtual objects, their placement, and composition for creation and communication. The biggest weakness, which also became apparent within the study, is the ability to represent behavior. Alliterations and steps in the user story have to be done manually.

Especially the first two phases show the difference for the creation of navigation concepts in an automotive context besides the advantages of spatial and tangible AR prototyping environments known so far in the literature. The structure and sequence developer show that the mental models and the language used to describe the user story significantly influence the requirements of a prototyping environment. For example, the structural descriptions showed that increased

knowledge of the vehicle's environment and possible occurring events in the environment is needed. On the other hand, the sequential descriptions showed the importance of knowledge about the vehicle's behavior, the components of navigation, procedures, and routines of the concept and the driver.

The analysis of the participants' description and approach supports the explicit multidimensional complexity of the automotive context of the generated AR application. The narrowing of the task helps this. The participants created their concept for only one maneuver. The processes would be significantly more complex in the context of an entire navigation route with multiple maneuvers and possible events in the process.

The traffic elements provided in the study, such as the crosswalk or the traffic light, were in all cases used either as anchors for virtual objects or as sources of information by the participants. This shows that the requirement for an immersive authoring tool is not only in the functional support of the concept developer, such as creating virtual objects or the description of animations and behavior. In particular, knowledge about the existence and function of the natural world is essential. Specifically, this means that the concept developer wants to know the state of the traffic light and interpret this information in his concept and present it to the driver in an appropriate way in the AR HUD.

Subdivision of the study into several phases may have influenced the participants by providing a thought pattern. Still, it helped structure the questions regarding the research question and all relevant items. Generally, the study was held according to the fixed standardized scheme, where the participants first had time for themselves, bouncing around some ideas. Afterwards, they were interviewed by the study director. The interview questions were the same with every participant, but subquestions in dialogue differed according to what each participant said before.

Covid-19 and the consequent limitations lead to several hurdles and problems in conducting the study. It was planned to observe and interview the participants in their presence. In the end, the study had to be moved to each participant's home, sending the set up to the participants and the study director conducting the study remotely. Next to technical issues, distractions at the participants' homes may have led to disruptions and delays. This also reflects in the evaluation, when some parts of the recording were incomprehensible and impossible to transcribe. Moreover, several participants had to back out due to Covid-19.

7 Conclusion

Domain-specific analysis for an AR authoring tool can add several complexity levels to the general problem of developing AR for in-vehicle use:

- Complexity of technology of the vehicle and suitable simulation environments.
- Complexity of traffic and other entities like real world elements and their functions (e.g. traffic lights) in relation to the vehicle.
- Complexity of behavior that can be addressed to own vehicle, other road users and the virtual objects.

A schematic representation of the complexity levels are shown in the Fig. 7.

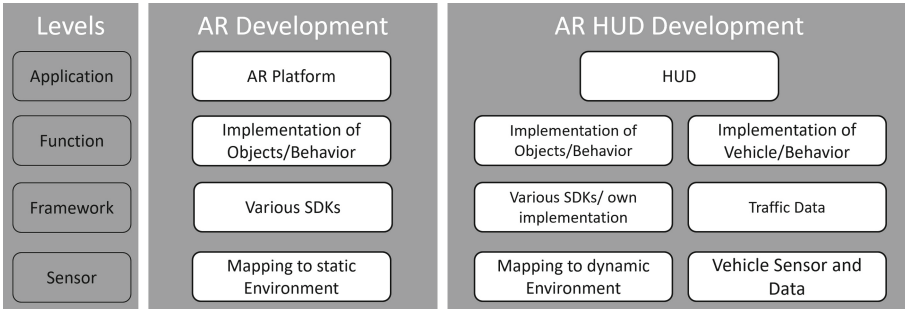


Fig. 7. Different complexities in developing AR- and AR HUD applications

Our guiding research questions have helped us understand the issues and dimensions involved in developing AR concepts for vehicle drivers. On the one hand, these are the consideration of the levels of complexity within the development and the inclusion of the individual aspects of the mobility domain that are not known in AR development for concepts. Thus, extensive knowledge about spatial and flow-oriented contexts is necessary, and the interaction with entities from traffic, environment and virtual objects is an essential requirement for creating AR concepts for vehicle drivers.

Because of the context explained so far, the complexities shown, and the wishes expressed by the concept developers, it is confirmed that a high fidelity and immersive authoring tool is beneficial for high ecological validity. Here all rules and information of the automotive mobility domain must be included. In addition to these rules, a multi-user evaluation in the target platform of the AR HUD is helpful.

A lo-fi prototyping environment leads to lower quality of developed AR concepts for vehicle drivers, higher production costs in terms of late changes in process, and a lack of understanding of the concept idea between developer, presenter, and customer.

One exemplary use case could be a rule-based development environment for a navigation concept that is related to a traffic light. Based on the traffic light's state, the concept would change its color to the represented status of the traffic light. This rule-based development should provide all necessary demands from ideation to presentation and speak the concept developer's language to achieve the best experience.

In the future, we plan to evaluate an immersive authoring tool for AR concepts for vehicle drivers to investigate the development process and the acceptance of such a development environment of a concept developer.

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competing conflict of interest. The results, opinions and conclusions expressed in this publication are not necessarily those of Volkswagen AG.

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